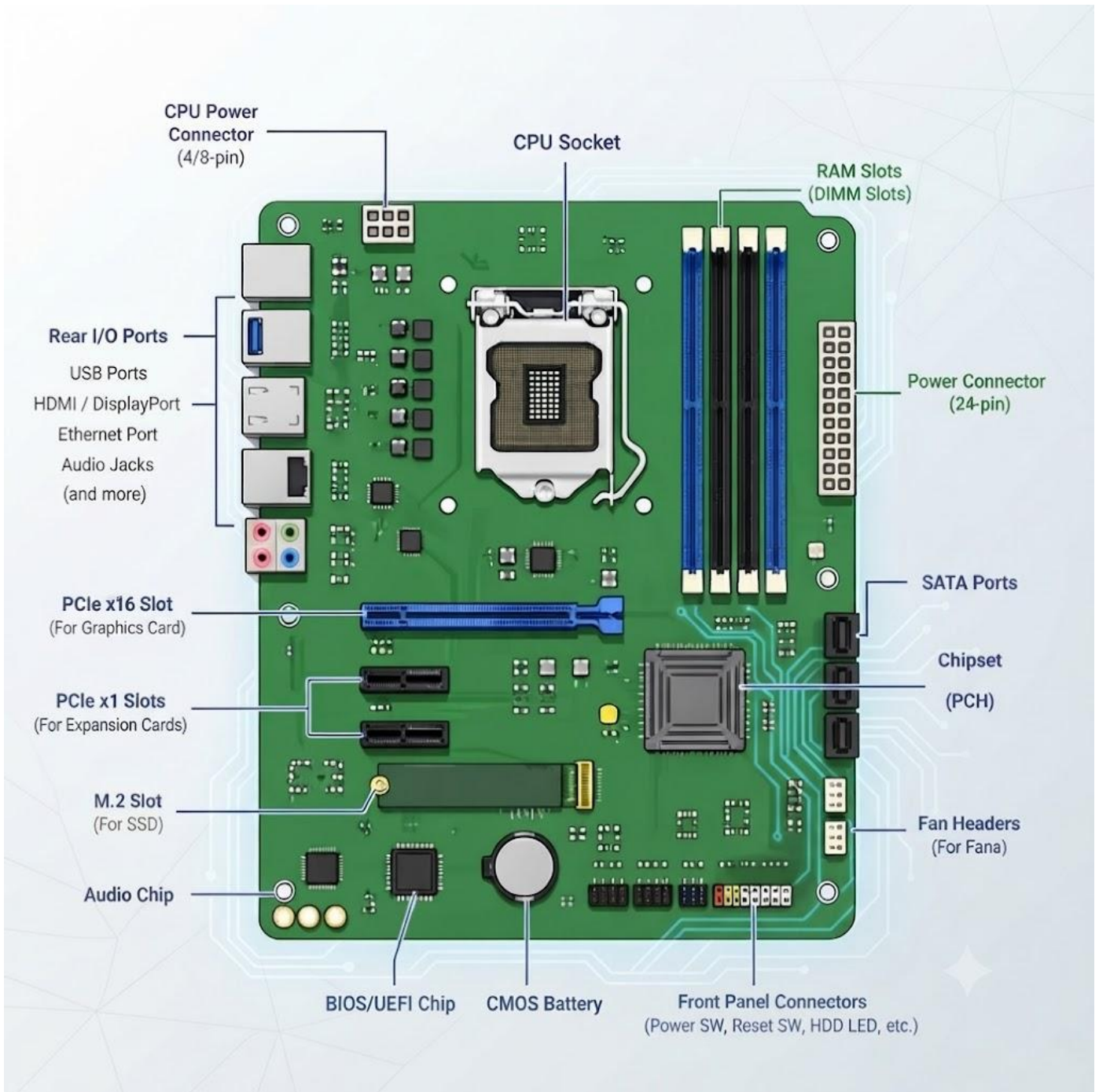


Types of motherboards and CPU socket compatibility

1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



1. **CPU Socket**: Where the central processing unit is installed
2. **RAM Slots (DIMM Slots)**: Slots used to install system memory.

Types of motherboards and CPU socket compatibility

3. **PCIe x16 Slot:** High-speed expansion slot typically used for dedicated graphics cards.
4. **24-Pin ATX Power Connector:** Supplies main electrical power from the power supply unit (PSU) to the motherboard.
5. **8-Pin ATX 12V Power Connector:** Delivers dedicated power directly to the CPU.
6. **SATA Ports:** Used to connect traditional mechanical hard drives (HDDs) and 2.5-inch solid-state drives (SSDs).
7. **M.2 NVMe Slot:** A high-speed slot for modern solid-state drives mounted flat against the board.
8. **Rear I/O Ports:** Houses external ports such as USB ports, audio jacks, Ethernet (LAN), and video outputs.

2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

➤ **Socket AM4 (AMD)**

- Compatible CPU: AMD Ryzen 7 5800X
- Supported Generations: AMD Ryzen 1000 through 5000 series desktop processors.

➤ **Socket LGA 1700 (Intel)**

- Compatible CPU: Intel Core i5-13600K
- Supported Generations: Intel 12th Gen (Alder Lake), 13th Gen (Raptor Lake), and 14th Gen (Raptor Lake Refresh) desktop processors.

➤ **Socket AM5 (AMD)**

- Compatible CPU: AMD Ryzen 7 7800X3D
- Supported Generations: AMD Ryzen 7000, 8000G, and 9000 series desktop processors.

Types of motherboards and CPU socket compatibility

3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

- **The CPU :** Does all the thinking, runs software commands, and calculates the data. The CPU acts as the primary "brain" of the computer by calculating data and executing software instructions. It decides what tasks need to be done and tells the other hardware components how to operate.
- **The Chipset :** Manages the communication paths, directing data back and forth between the CPU and parts like storage drives, USB ports, internet cards, and sound chips.

❖ What Happens When You Open & Play Spotify:

- You Click the Icon: Your mouse sends a click signal through a USB port. The chipset receives it and passes it along to the CPU.
- Loading the App: The CPU tells the chipset to grab Spotify's files from your storage drive (SSD) and copy them into your RAM.
- Opening the Screen: The CPU reads those files from RAM and tells your screen to display the Spotify window.
- Streaming the Track: When you press play, the CPU tells the chipset to download the music stream using your Wi-Fi or Ethernet card.
- Playing Sound: The CPU unpacks the music file into audio data and sends it through the chipset to your headphone or speaker jack so you can hear it.

Types of motherboards and CPU socket compatibility

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

